

SIXTH SEMESTER DIPLOMA EXAMINATION IN CIVIL ENGINEERING – MARCH 2013

WATER RESOURCES ENGINEERING

Time : 3 hours

(Maximum marks : 100)

PART – A (Maximum marks : 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. What are the different sources of water?
2. Define Precipitation.
3. What is Water Conservation?
4. Distinguish between Confined and Unconfined aquifer.
5. Define drought.

(5 x 2 = 10)

PART – B (Maximum marks : 30)

II. Answer any five questions. Each question carries 6 marks.

1. Explain Water quality management.
2. What are the factors affecting run off.
3. Explain briefly the Water budget.
4. What are the methods adopted for improving percolation of water?
5. What is a Check dam ? What are the functions of a check dam?
6. Explain Pitched island with the aid of a sketch.
7. State Darcy's law. Explain steady radial flow to a well.

(5 x 6 = 30)

PART – C (Maximum marks : 60)

(Answer one full question from each module. Each question carries 15 marks)

MODULE - I

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| III. (a) Explain the need for a National water policy. | 7 |
| (b) Discuss the rainfall distribution in India. | 8 |

OR

- IV. (a) What are the planning arrangements for a flood control project. 8
(b) Explain fresh water management. 7

MODULE – II

- V. (a) Describe with a neat sketch the Hydrologic cycle. 8
(b) What is evapotranspiration? Explain the inflow outflow method for evapotranspiration. 7

OR

- VI. (a) List out the different types of self recording rain gauges. Explain the working of any two. 8
(b) What is infiltration? Explain how infiltration is measured by double ring Infiltrometer. 7

MODULE – III

- VII. (a) Explain the need for water conservation and management. 7
(b) Explain the roof water collection and storage system. 8

OR

- VIII. (a) Explain the role of forest in conserving water. 7
(b) Explain briefly Contourbunding and Terracing method of ground recharge. 8

MODULE – IV

- IX. (a) What is a Guide bank? Sketch a suitable cross-section of a guide bank as used In river training works. 7
(b) Explain the different types of Tube wells. 8

OR

- X (a) Draw the section of a Groyne and describe in brief various types of groyne used for river training. 8
(b) Discuss briefly the various types of reservoirs. 7
